Probability Theory and Statistics

Sample Midterm 2
Solutions

1. Two numbers are chosen independently and uniformly at random from the interval $[0, 1]$. What is the probability that the sum of the two numbers is greater than twice the absolute difference of the larger and smaller number?

(2 pts) The random experiment is equivalent to selecting a random point in the square $[0,1] \times [0,1]$.

(1 pt) This is the sample space Ω.

(2 pts) Denote the random point by (x, y). The favorable region consists of those points for which either $x > y$ and $x + y > 2(x - y)$, or $y > x$ and $x + y > 2(y - x)$.

(2 pts) These inequalities can be rewritten into a form suitable for drawing the figure / computing the favorable area: $x > y > x/3$ or $y > x > y/3$ (or equivalently, expressing x as a function of y).

(2 pts) Thus the favorable region is the part of the square between the lines $y = x/3$ and $y = x$, and between $y = x$ and $y = 3x$.

(3 pts) A correct and well-labelled figure.

(3 pts) In a geometric probability space, $\mathbb{P} = \dfrac{T_{\text{fav}}}{T_\Omega}$.

(1 pt) $T_{\text{fav}} = 1 - T_{\text{unfav}}$.

(1 pt) $T_\Omega = 1$.

(2 pts) $T_{\text{unfav}} = 2 \times \dfrac{1 \cdot (1/3)}{2} = \dfrac{1}{3}$ (since we are summing the areas of two right triangles).

(0 pt) Therefore $T_{\text{fav}} = 2/3$.

(1 pt) Hence the required probability is $\boxed{\mathbb{P} = 2/3.}$

Remarks: If the student computes the unfavorable area first instead of the favorable one, or expresses the inequalities differently but correctly, full credit is given. Even if the figure is missing, but it is clear from the text or calculations that the student computed the correct geometric areas, points should be awarded for the figure step as well.

2. In a forest there are two kinds of felines in large numbers: domestic cats and wildcats. Wildcats make up 10% of all felines. All wildcats are tabby, and among domestic cats every fifth is tabby on average.

   (a) If one feline is selected at random, what is the probability that it is tabby?

   (b) If a randomly spotted feline is tabby, but we cannot tell from a distance whether it is domestic or wild, what is the probability that it is domestic?

   (c) Are the following events independent?

$$A = \{\text{the selected feline is } \textit{not} \text{ tabby}\}, \qquad B = \{\text{the selected feline is domestic}\}.$$

(a) (9 pts)

(1 pt) The required probability is $\mathbb{P}(A)$ (or the student expresses it in their own notation).

(1 pt) $\mathbb{P}(B) = 9/10$,

(1 pt) hence $\mathbb{P}(\bar{B}) = 1/10$.

(1 pt) $\mathbb{P}(A \mid B) = 1/5$,

(1 pt) and $\mathbb{P}(A \mid \bar{B}) = 1$.

(1 pt) Since B and $\bar{B}$ form a complete system of disjoint events,

(1 pt) by the law of total probability,

$$\mathbb{P}(A) = \mathbb{P}(B)\mathbb{P}(A \mid B) + \mathbb{P}(\bar{B})\mathbb{P}(A \mid \bar{B})$$

(1 pt) $= \frac{9}{10} \cdot \frac{1}{5} + \frac{1}{10} \cdot 1 = \frac{7}{25} = 0.28.$

(b) (5 pts)

(1 pt) The required conditional probability is $\mathbb{P}(B \mid A)$.

(1 pt) By Bayes' theorem,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

(2 pts) Substituting:

$$\mathbb{P}(B \mid A) = \frac{(1/5) \cdot (9/10)}{(1/5) \cdot (9/10) + 1 \cdot (1/10)} = \frac{9}{14} \approx 0.643.$$

(c) (6 pts)

We check whether A and B are independent.

(1 pt) Independence would mean $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

(1 pt) $\mathbb{P}(A) = 1 - \mathbb{P}(\bar{A}) = 18/25$.

(1 pt) $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \frac{4}{5} \cdot \frac{9}{10} = \frac{18}{25} \neq \frac{81}{125} = \mathbb{P}(A)\mathbb{P}(B)$.

(1 pt) Therefore they are not independent.

Alternative reasoning: Independence is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ or to $\mathbb{P}(B \mid A) = \mathbb{P}(B)$. Since $\mathbb{P}(A \mid B) = 4/5 \neq 18/25$, and $\mathbb{P}(B \mid A) = 1 \neq 9/10$, A and B are not independent.

3. Anna, Bea, and Csilla play with a 32-card Hungarian deck. After shuffling, each receives 8 cards. The deck contains exactly 8 red cards.

   (a) What is the probability that Anna has no red cards?

   (b) What is the probability that neither Bea nor Csilla has a red card?

   (c) What is the probability that all three players have at least one red card?

(a) (5 pts)

(1 pt) Let A, B, and C denote the events that Anna, Bea, and Csilla, respectively, have at least one red card. We are looking for $\mathbb{P}(\bar{A})$.

(1 pt) Anna receives 8 cards from the 24 non-red ones, out of the 32 total cards:

$$\mathbb{P}(\bar{A}) = \frac{\binom{24}{8}}{\binom{32}{8}} \approx 0.0699.$$

(b) (5 pts)

(1 pt) We are looking for $\mathbb{P}(\bar{B} \cap \bar{C})$.

(1 pt) Bea draws 8 cards from the 24 non-red cards, Csilla draws 8 from the remaining 16 non-red ones:

$$\mathbb{P}(\bar{B} \cap \bar{C}) = \frac{\binom{24}{8}\binom{16}{8}}{\binom{32}{8}\binom{24}{8}} = \frac{16!\,24!}{32!\,8!} \approx 0.0012.$$

(c) (10 pts)

(1 pt) The required probability is $\mathbb{P}(A \cap B \cap C)$.

(1 pt) By the complement rule, $\mathbb{P}(A \cap B \cap C) = 1 - \mathbb{P}(\bar{A} \cup \bar{B} \cup \bar{C})$.

(1 pt) By symmetry, $\mathbb{P}(\bar{A}) = \mathbb{P}(\bar{B}) = \mathbb{P}(\bar{C})$ and $\mathbb{P}(\bar{A} \cap \bar{B}) = \mathbb{P}(\bar{A} \cap \bar{C}) = \mathbb{P}(\bar{B} \cap \bar{C})$.

(1 pt) By Poincaré's formula,

$$\mathbb{P}(\bar{A} \cup \bar{B} \cup \bar{C}) = 3\mathbb{P}(\bar{A}) - 3\mathbb{P}(\bar{A} \cap \bar{B}) + \mathbb{P}(\bar{A} \cap \bar{B} \cap \bar{C}).$$

(2 pts) The event $\bar{A} \cap \bar{B} \cap \bar{C}$ means that none of the players has any red cards, i.e. all 8 red cards remain among the 8 un-dealt cards:

$$\mathbb{P}(\bar{A} \cap \bar{B} \cap \bar{C}) = \frac{1}{\binom{32}{8}} = \frac{24!\,8!}{32!} \approx 9.5 \cdot 10^{-6}.$$

(1 pt) Substituting the values,

$$\mathbb{P}(A \cap B \cap C) = 1 - (3\mathbb{P}(\bar{A}) - 3\mathbb{P}(\bar{B} \cap \bar{C}) + \mathbb{P}(\bar{A} \cap \bar{B} \cap \bar{C})) \approx 0.7939.$$

4. A university committee of nine members must choose between two program options, A and B. Three members definitely vote for A, two definitely for B, and the remaining four are undecided. Each undecided member flips a fair coin independently; if the result is *heads*, they vote for A, and if *tails*, for B. Let X denote the number of votes for A.

   (a) What is the probability that one of the two options wins by more than two votes?

   (b) What is the distribution (and parameters) of the random variable $X - 3$?

   (c) Let Y denote the number of votes for option B. Determine the joint distribution of (X, Y) and their marginal distributions (preferably in tabular form).

(a) (7 pts)

(1 pt) $\mathbb{P}(\text{an option wins by more than 2 votes}) = \mathbb{P}(A \text{ wins by } > 2) + \mathbb{P}(B \text{ wins by } > 2)$.

(1 pt) A wins by more than two votes if A receives exactly 6 or 7 votes, i.e. $\mathbb{P}(X = 6) + \mathbb{P}(X = 7)$.

(2 pts)

$$\mathbb{P}(X = 6) + \mathbb{P}(X = 7) = \binom{4}{3}\left(\frac{1}{2}\right)^4 + \binom{4}{4}\left(\frac{1}{2}\right)^4 = \frac{5}{16}.$$

(1 pt) Similarly, B wins by more than two votes if A receives only 3 votes, i.e. $\mathbb{P}(X = 3)$.

$$\mathbb{P}(X = 3) = \binom{4}{0}\left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

(2 pts) Hence the required probability is

$$\boxed{\mathbb{P} = \frac{3}{8} = 0.375.}$$

(b) (3 pts)

(2 pts) $X - 3 \sim \text{Bin}(4,\ 1/2)$.

(1 pt) Justification: the four undecided members perform four independent trials, each resulting in a vote for A (a "success") with probability $1/2$.

(c) (10 pts)

Let Y denote the number of votes for B. Clearly, $X + Y = 9$ with probability 1.

The joint distribution and marginals are:

$Y\backslash X$	3	4	5	6	7	$p_Y(l)$
2	0	0	0	0	$\frac{1}{16}$	$\frac{1}{16}$
3	0	0	0	$\frac{1}{4}$	0	$\frac{1}{4}$
4	0	0	$\frac{3}{8}$	0	0	$\frac{3}{8}$
5	0	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
6	$\frac{1}{16}$	0	0	0	0	$\frac{1}{16}$
$p_X(k)$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{16}$	1

That is, the only nonzero probabilities occur when $X+Y = 9$. The marginals follow by summing over rows/columns, giving p_X and p_Y as in the last line/column.

5. A continuous random variable X has density function

$$f_X(x) = \begin{cases} \dfrac{c}{x^2}, & x > 1, \\ 0, & \text{otherwise}, \end{cases}$$

where $c > 0$ is an appropriate constant.

(a) Determine the value of c.

(b) Define a discrete random variable Y as follows:

$$Y = \begin{cases} 3, & \text{if } X < 3, \\ 0, & \text{if } 3 \le X < 4, \\ 4, & \text{otherwise.} \end{cases}$$

Find $\mathbb{P}(Y = k)$ for all k with positive probability.

(c) Compute the expectation and standard deviation of Y.

(a) (6 pts)

(1 pt) f_X is a density if it is nonnegative and integrates to 1.

(2 pts)

$$1 = \int_{-\infty}^{\infty} f_X(x)\,dx = \int_1^{\infty} \frac{c}{x^2}\,dx = c\,[-1/x]_1^{\infty} = c.$$

(1 pt) Hence $\boxed{c = 1.}$

(b) (7 pts)

(1 pt) $\text{Ran}(Y) = \{0, 3, 4\}$.

(2 pts)

$$\mathbb{P}(Y = 3) = \mathbb{P}(1 < X < 3) = \int_1^3 \frac{1}{x^2}\,dx = [-1/x]_1^3 = 1 - \tfrac{1}{3} = \tfrac{2}{3}.$$

(2 pts)

$$\mathbb{P}(Y = 0) = \mathbb{P}(3 \le X < 4) = \int_3^4 \frac{1}{x^2}\,dx = [-1/x]_3^4 = \tfrac{1}{3} - \tfrac{1}{4} = \tfrac{1}{12}.$$

(2 pts)

$$\mathbb{P}(Y = 4) = \mathbb{P}(X \ge 4) = \int_4^{\infty} \frac{1}{x^2}\,dx = [-1/x]_4^{\infty} = \tfrac{1}{4}.$$

(c) (7 pts)

(2 pts) $\mathbb{E}(Y) = 3 \cdot \frac{2}{3} + 4 \cdot \frac{1}{4} = 3.$
(2 pts) $\mathbb{E}(Y^2) = 9 \cdot \frac{2}{3} + 16 \cdot \frac{1}{4} = 10.$
(2 pts) $\mathbb{D}^2(Y) = 10 - 3^2 = 1.$
(1 pt) $\boxed{\mathbb{D}(Y) = 1.}$

6. Let X be a random variable with distribution function

$$F_X(x) = e^{-e^{-x}} \qquad (e \approx 2.71).$$

(a) Prove that there exists $\varepsilon > 0$ such that $\mathbb{P}(X \le \varepsilon) < \frac{1}{2}$.
(b) Let $Y = e^{-X}$. Determine the density function and standard deviation of Y.

(a) (6 pts)

(1 pt) $\mathbb{P}(X < 0) = F_X(0) = e^{-e^0} = e^{-1} \approx 0.3679 < 1/2.$
(1 pt) F_X is continuous, since it is a composition of continuous functions.
(2 pts) Hence, by continuity, there exists $\varepsilon > 0$ such that $F_X(\varepsilon) < 1/2$.
(2 pts) Consequently, $\mathbb{P}(X \le \varepsilon) = F_X(\varepsilon) < 1/2$.

(b) (14 pts)

(1 pt) To find the density of Y, we first determine its distribution function.
(1 pt) Range: $\mathrm{Ran}(Y) = (0, \infty)$.
(1 pt) For $x \le 0$, $F_Y(x) = 0$.
(3 pts) For $x > 0$:

$$F_Y(x) = \mathbb{P}(Y < x) = \mathbb{P}(e^{-X} < x) = \mathbb{P}(X > -\ln x) = 1 - F_X(-\ln x) = 1 - e^{-x}.$$

(2 pts) Hence $Y \sim \mathrm{Exp}(1)$.
(2 pts) The density is

$$f_Y(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

(2 pts) For the exponential distribution with $\lambda = 1$, $\mathbb{D}^2(Y) = 1$.
(1 pt) Thus $\boxed{\mathbb{D}(Y) = 1.}$